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Do ETFs Increase Volatility?

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1 Big picture

- **Question.** Do exchange-traded funds increase volatility in the stocks they hold?
- **Main mechanism.** ETFs provide very cheap, intraday, basket-level trading. This attracts short-horizon liquidity traders. ETF arbitrage then transmits ETF-level demand shocks into the underlying securities.
- **Core contrast.** ETFs could either reduce volatility by buffering demand shocks, or increase volatility by making underlying stocks inherit ETF demand shocks. The paper tests which force dominates.
- **Main empirical result.** Stocks with higher ETF ownership have higher daily volatility. The relation is positive in OLS and remains positive under a Russell-index reconstitution instrument.
- **Identification.** The paper exploits the Russell 1000/Russell 2000 cutoff. Switching from Russell 1000 to Russell 2000 mechanically raises ETF ownership; switching the other way lowers it.
- **Mechanism evidence.** ETF ownership attracts high-turnover traders to underlying stocks, ETF flows create same-day price pressure, and much of that price pressure reverses over subsequent days.
- **Noise versus information.** The reversal evidence and negative return autocorrelation suggest that the volatility increase is at least partly nonfundamental rather than simply faster price discovery.
- **Limits-to-arbitrage evidence.** The volatility effect is tied to ETF mispricing and is weaker when arbitrage is more costly, such as when bid-ask spreads or lending fees are high.
- **Asset-pricing implication.** Stocks with high ETF ownership earn a premium. A high-minus-low ETF ownership portfolio earns statistically significant monthly alphas of about 30–38 bps in the full sample and up to about 56 bps in later subsamples.
- **Takeaway.** ETF ownership is not purely a passive wrapper. Through arbitrage and liquidity trading, ETFs can add a new layer of demand shocks to individual stocks.

2 Data and measurement

2.1 Main sample and ownership measures

- The sample covers U.S. equity ETFs and U.S. stocks from January 2000 to December 2015.
- The main stock universes are S&P 500 stocks and Russell 3000 stocks.
- ETF ownership of stock i at time t is the fraction of stock i 's market capitalization held by ETFs.
- The paper also measures ownership by index mutual funds, active mutual funds, and hedge funds to separate ETF effects from general institutional ownership.
- ETF ownership rises sharply over the sample: in S&P 500 stocks, ETF ownership grows from about 0.14% of market capitalization in 2000 to about 7.05% in 2015.

Interpretation. ETF ownership is the treatment intensity. It measures how strongly stock prices are exposed to arbitrage trades triggered by ETF-level demand.

2.2 Volatility and liquidity variables

- Monthly stock-level volatility is computed from daily returns within the month.
- Daily intraday volatility is computed from high-frequency returns and is used in arbitrage-channel tests.
- Liquidity controls include inverse price, Amihud price impact, bid-ask spread, and turnover.

- Standard return predictors include market capitalization, book-to-market, past 12-month returns, and gross profitability.

Interpretation. The controls are designed to separate ETF ownership from size, liquidity, valuation, momentum, profitability, and preexisting volatility.

2.3 Investor trading horizon: churn ratio

The paper uses an Ancerno trade-level measure of how frequently institutions buy and sell the same security. For manager k , security i , and quarter q ,

$$\text{Churn Ratio}_{k,i,q} = 1 - \frac{|\$BuyTrades_{k,i,q} - \$SellTrades_{k,i,q}|}{\$BuyTrades_{k,i,q} + \$SellTrades_{k,i,q}}. \quad (1)$$

To incorporate repeated direction changes, the adjusted churn ratio is

$$\text{Adjusted Churn Ratio}_{k,i,q} = \text{Churn Ratio}_{k,i,q} \times \frac{N_{k,i,q} + 1}{2}, \quad (2)$$

where $N_{k,i,q}$ is the number of times the trade direction switches during the quarter.

Interpretation. ETFs should attract higher-churn investors if their low trading costs make them preferred vehicles for short-horizon trading.

2.4 ETF flows and stock-level exposure

ETF flows are measured at the ETF level and aggregated to the stock level according to ETF holdings:

$$Flows_{i,t} = \frac{\sum_{j=1}^J w_{i,j,t} \$Flows_{j,t}}{\$Volume_{i,t}}, \quad (3)$$

where $w_{i,j,t}$ is ETF j 's ownership weight in stock i and the denominator is the stock's recent trading volume.

Interpretation. If ETF flows are liquidity shocks, they should push stock prices in the same direction initially and then reverse.

2.5 Russell-index instrument

- The Russell 1000 contains the largest 1,000 firms; the Russell 2000 contains the next 2,000.
- Around the cutoff, stocks switching from Russell 1000 to Russell 2000 receive more ETF ownership because they become larger weights in Russell 2000-linked ETFs.
- Stocks switching from Russell 2000 to Russell 1000 experience a decline in ETF ownership.
- The paper uses the switch indicator as an instrument for ETF ownership in a local sample around the cutoff.

Interpretation. The index cutoff provides quasi-exogenous variation in ETF ownership that is less likely to be driven by volatility itself.

3 Models and empirical specifications

3.1 Liquidity-shock propagation

The paper's central mechanism can be summarized as

ETF liquidity shock $\rightarrow$ ETF price moves $\rightarrow$ ETF arbitrage $\rightarrow$ basket stocks trade $\rightarrow$ stock prices inherit shock.

Interpretation. The ETF is not merely a passive ownership vehicle. Its trading can move the underlying securities through the arbitrage process.

3.2 Baseline OLS volatility regression

The baseline monthly specification is

$$Volatility_{i,t} = \alpha_i + \delta_t + \beta ETFOwnership_{i,t-1} + \Gamma' X_{i,t-1} + \varepsilon_{i,t}, \quad (4)$$

where α_i are stock fixed effects, δ_t are month fixed effects, and X includes size, liquidity, return predictors, lagged volatility, and other institutional ownership.

Interpretation. A positive β means that a stock becomes more volatile when a larger fraction of it is held by ETFs, after comparing the stock to itself and controlling for aggregate time shocks.

3.3 Russell IV design

The first stage is

$$ETFOwnership_{i,t} = \alpha_i + \delta_t + \pi Switch_i + \Gamma' X_{i,t-1} + u_{i,t}. \quad (5)$$

The second stage is

$$Volatility_{i,t} = \alpha_i + \delta_t + \beta \widehat{ETFOwnership}_{i,t-1} + \Gamma' X_{i,t-1} + \varepsilon_{i,t}. \quad (6)$$

Interpretation. The instrument changes ETF ownership mechanically through index membership. The exclusion restriction is that index switching affects volatility only through ETF ownership, conditional on controls and local rank polynomials.

3.4 ETF flows and return reversal

The flow-return tests estimate how stock returns respond to ETF-induced stock-level flows at different horizons:

$$r_{i,t+h} = \alpha_i + \delta_t + \beta_h Flows_{i,t} + \Gamma' X_{i,t} + \varepsilon_{i,t+h}. \quad (7)$$

Interpretation. If $\beta_0 > 0$ and later $\beta_h < 0$, ETF flows create temporary price pressure rather than purely revealing fundamentals.

3.5 Variance-ratio tests

The paper studies whether ETF ownership makes stock prices deviate from a random walk by using variance-ratio outcomes.

- A random walk implies a variance ratio close to one.
- Negative autocorrelation implies a variance ratio below one.
- The paper uses both the absolute deviation from one and the signed variance ratio.

Interpretation. Higher ETF ownership should raise mean reversion if ETFs add nonfundamental liquidity shocks.

3.6 Limits-to-arbitrage regression

The arbitrage-channel tests interact stock-level ETF mispricing with arbitrage-cost proxies:

$$IntradayVol_{i,t} = \alpha_i + \delta_t + \beta |Mispricing|_{i,t-1} + \theta |Mispricing|_{i,t-1} \times HighCost_{i,t-1} + \Gamma' X_{i,t-1} + \varepsilon_{i,t}. \quad (8)$$

Interpretation. If ETF arbitrage transmits shocks, ETF mispricing should predict next-day stock volatility. If arbitrage is costly, the effect should be weaker; thus $\theta < 0$ for high bid-ask spreads or high lending fees.

3.7 ETF ownership premium

The asset-pricing tests form monthly quintile portfolios sorted on lagged ETF ownership and estimate factor-model alphas for

$$R_t^{HighETF} - R_t^{LowETF}.$$

The paper also runs Fama–MacBeth regressions of abnormal returns on lagged ETF ownership and contemporaneous ETF flows.

Interpretation. If ETF-induced volatility creates systematic risk, investors require compensation for holding high-ETF-ownership stocks.

4 Identification and empirical design

4.1 What OLS shows

- OLS provides a broad full-sample relation between ETF ownership and volatility.
- Stock fixed effects absorb time-invariant stock characteristics.
- Month fixed effects absorb market-wide volatility shocks.
- Controls account for liquidity, size, valuation, profitability, momentum, lagged volatility, and other institutional ownership.

Interpretation. OLS is informative but not definitive because ETF ownership may be endogenous to the stocks that ETF sponsors choose to hold and to investor demand for certain stocks.

4.2 Why Russell reconstitution helps

- The Russell cutoff changes index-linked ETF demand mechanically.
- Around the cutoff, firms are similar in size but receive different index weights and therefore different ETF ownership.
- The paper uses bandwidths from 100 to 500 stocks around the cutoff.
- It focuses on the pre-banding-rule period when the cutoff assignment was cleaner.

Interpretation. This design moves the analysis closer to causal inference by using externally generated variation in ETF ownership.

4.3 Threats to exclusion and how the paper responds

- **Visibility/information production.** Switching indexes might change analyst coverage or media attention. Prior evidence and the paper’s reversal tests make this less likely as the full explanation.
- **Other institutional ownership.** Index, active, and hedge fund ownership may also move around the Russell cutoff. The paper controls for them and tests ratios of ETF ownership to other ownership types.
- **Time-varying ETF importance.** The instrument should be stronger when aggregate ETF ownership is higher. Table V Panel C supports this implication.
- **Local nature.** IV magnitudes apply to stocks near the Russell cutoff and may not generalize to all stocks.

Interpretation. The paper’s causal case is strongest when the evidence is viewed as a bundle: OLS, Russell IV, return reversals, variance ratios, and arbitrage-cost interactions all point in the same direction.

4.4 What the paper cannot fully prove

- The Russell design identifies a local treatment effect, not necessarily a universal market-wide effect.
- ETF flows and arbitrage are measured with proxies; actual AP inventory and all arbitrage trades are not directly observed.
- Return reversals support a nonfundamental interpretation, but price discovery and liquidity trading can coexist.
- The expected-return premium may reflect ETF-induced risk, but asset-pricing tests cannot fully rule out every omitted risk characteristic.

Interpretation. The paper is strongest on the claim that ETF ownership is associated with, and plausibly causes, higher volatility; the exact decomposition between noise, information, and risk premia is harder.

5 Tables and figures

5.1 Table I: ETF ownership statistics

Read:

- ETF ownership rises dramatically from 2000 to 2015.
- In S&P 500 stocks, ETF ownership increases from 0.14% of market capitalization in 2000 to 7.05% in 2015.
- In Russell 3000 stocks, ETF ownership rises from 0.14% to 7.48%.
- ETF ownership grows much faster than index mutual fund ownership.

Economic message. ETF ownership becomes economically important during the sample, creating the conditions for ETFs to transmit demand shocks to stocks.

Table I
ETF Ownership Statistics

The table presents descriptive statistics for stock ownership by ETFs, index mutual funds, and active mutual funds for the S&P 500 and Russell 3000 universes. For each year, across months and stocks, we average the number of institutions and the percentage of each stock owned by an institution type. We also report cumulative assets owned by these institutions in the stocks in our sample, as reported in Thomson-Reuters Mutual Fund Ownership data.

S&P 500									
Year	ETFs			Index Funds			Active Funds		
	Number of Funds	Percentage of Market Cap	Market Cap (in Dollars)	Number of Funds	Percentage of Market Cap	Market Cap (in Dollars)	Number of Funds	Percentage of Market Cap	Market Cap (in Dollars)
2000	4.5	0.14%	\$16.6	111.6	2.61%	\$318.6	240.5	15.22%	\$1,872.1
2001	12.6	0.19%	\$18.0	126.7	2.65%	\$269.7	277.8	15.65%	\$1,616.9
2002	14.2	0.46%	\$39.6	137.6	2.76%	\$239.9	288.4	16.07%	\$1,416.2
2003	15.1	0.69%	\$62.4	141.7	2.92%	\$262.9	286.0	16.10%	\$1,471.9
2004	18.4	0.86%	\$80.6	142.4	3.23%	\$141.2	284.3	16.83%	\$1,264.4
2005	20.2	1.16%	\$129.5	139.3	3.44%	\$383.8	287.6	17.55%	\$1,980.4
2006	21.5	1.29%	\$157.2	130.4	3.64%	\$444.4	291.2	18.10%	\$2,231.2
2007	38.5	2.50%	\$336.2	128.6	3.04%	\$407.8	307.5	19.66%	\$2,658.5
2008	54.2	4.16%	\$450.6	131.1	2.28%	\$257.5	345.2	21.18%	\$2,213.3
2009	56.0	4.66%	\$412.1	125.8	2.51%	\$223.0	357.2	21.56%	\$1,913.9
2010	54.7	4.77%	\$506.6	113.7	2.60%	\$275.9	337.2	21.44%	\$2,281.3
2011	54.9	5.21%	\$604.6	108.6	2.61%	\$303.2	338.6	21.72%	\$2,527.8
2012	51.5	5.58%	\$717.6	99.9	2.73%	\$351.2	399.8	20.90%	\$2,695.8
2013	53.4	6.30%	\$958.0	97.9	2.87%	\$457.2	384.5	21.41%	\$3,265.6
2014	47.7	6.47%	\$1,140.7	91.7	2.89%	\$512.0	360.1	20.96%	\$3,697.0
2015	48.3	7.05%	\$1,214.2	92.9	3.20%	\$550.9	257.3	21.70%	\$3,741.8

(Continued)

Table I—Continued
Russell 3000

Year	ETFs			Index Funds			Active Funds		
	Number of Funds	Percentage of Market Cap	Market Cap (in Dollars)	Number of Funds	Percentage of Market Cap	Market Cap (in Dollars)	Number of Funds	Percentage of Market Cap	Market Cap (in Dollars)
2000	2.2	0.12%	\$17.5	40.4	2.33%	\$343.1	80.3	16.44%	\$2,456.3
2001	7.5	0.19%	\$23.3	46.6	2.49%	\$298.7	92.7	16.59%	\$2,006.3
2002	8.1	0.48%	\$48.9	51.5	2.65%	\$276.5	97.4	17.09%	\$1,773.8
2003	8.7	0.73%	\$80.6	56.3	2.83%	\$310.4	98.4	17.11%	\$1,882.0
2004	10.6	0.93%	\$121.2	57.2	3.15%	\$413.4	102.6	17.59%	\$2,361.5
2005	11.5	1.24%	\$173.2	56.2	3.38%	\$473.3	105.5	18.74%	\$2,637.3
2006	12.5	1.39%	\$210.8	55.4	3.66%	\$555.7	105.1	19.14%	\$2,932.8
2007	22.4	2.65%	\$445.4	55.3	3.09%	\$518.2	124.8	21.02%	\$3,563.3
2008	31.0	4.34%	\$555.8	57.9	2.26%	\$291.7	144.0	22.54%	\$2,941.7
2009	31.2	4.95%	\$534.2	58.5	2.51%	\$271.3	147.9	23.07%	\$2,496.0
2010	30.1	5.14%	\$677.5	52.2	2.56%	\$326.5	139.8	23.07%	\$3,050.8
2011	30.0	5.56%	\$803.4	51.4	2.61%	\$375.5	134.2	23.28%	\$3,382.7
2012	28.2	5.96%	\$942.1	49.1	2.71%	\$427.0	122.9	22.49%	\$3,572.6
2013	29.7	6.70%	\$1,267.7	47.9	2.86%	\$537.2	119.2	22.89%	\$4,370.6
2014	27.6	6.83%	\$1,502.4	45.4	2.89%	\$622.2	109.8	22.21%	\$4,917.8
2015	26.3	7.48%	\$1,611.3	44.9	3.14%	\$673.6	109.4	22.82%	\$4,974.1

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(a) S&P 500 ETF ownership

(b) Russell 3000 ETF ownership

5.2 Table II: Summary statistics

Read:

- The paper reports stock-month variables, correlations, daily variables used in return regressions, and Ancerno trading variables.
- ETF ownership is small on average but highly dispersed, giving useful cross-sectional variation.
- ETF flows are scaled by stock trading volume, making the flow measure comparable across firms.
- Ancerno data allow the paper to measure institutional trading intensity and price impact directly.

Economic message. Table II defines the empirical objects used throughout the paper: ownership, volatility, flows, liquidity, and investor turnover.

Table II
Summary Statistics

The table presents summary statistics for the variables used in the study. Panel A reports summary statistics for the stock-month sample. Panel B reports correlations for the same sample, and Panel C reports summary statistics for the stock-day sample. Panel D reports summary statistics for the Ancerno transaction-level data. The samples cover the period January 2000 to December 2015.

Panel A: Monthly Sample

	S&P 500					
	<i>N</i>	Mean	<i>SD</i>	Min	Median	Max
Daily stock volatility (%)	81,243	2.010	1.400	0.581	1.690	11.000
Skewness	81,243	0.109	2.490	-8.900	0.114	8.900
LogAbsVR 5 days(5*1 day))	81,243	-1.810	0.895	-3.850	-1.600	-0.582
ETF ownership	81,243	0.026	0.019	0.000	0.023	0.112
Index fund ownership	81,243	0.062	0.020	0.004	0.061	0.120
Active fund ownership	81,243	0.176	0.063	0.008	0.173	0.354
Hedge fund ownership	81,243	0.034	0.045	0.000	0.019	0.354
log(Mktcap (\$m))	81,243	9.369	1.060	6.230	9.300	11.400
lPrice	81,243	0.009	0.040	0.004	0.009	0.556
Amihud ratio	81,243	0.000	0.001	0.000	0.000	0.050
Bid-ask spread (%)	81,243	0.297	0.462	0.000	0.048	3.700
Book-to-market	81,243	0.524	0.385	0.038	0.420	2.260
Past 12-month returns	81,243	0.113	0.366	-0.823	0.100	2.180
Gross profitability	81,243	0.301	0.224	-0.399	0.262	1.120

(Continued)

(a) Panel A summary statistics, part 1

Table II—Continued

Panel B: Correlations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)		
Daily stock volatility (%)	(1)	1.000													
Skewness	(2)	-0.004	1.000												
LogAbsVR 5 days(5*1 day))	(3)	0.027	0.000	1.000											
ETF ownership (%)	(4)	-0.074	-0.021	-0.022	1.000										
Index fund ownership (%)	(5)	-0.092	-0.007	-0.060	0.355	1.000									
Active fund ownership (%)	(6)	-0.102	-0.004	-0.097	0.160	0.428	1.000								
Hedge fund ownership (%)	(7)	0.071	0.008	-0.025	0.055	-0.005	-0.052	1.000							
log(Mktcap (\$m))	(8)	-0.343	-0.001	-0.083	0.041	0.325	0.382	-0.147	1.000						
lPrice	(9)	0.402	0.001	0.040	-0.074	-0.169	-0.287	0.084	-0.459	1.000					
Amihud ratio	(10)	0.112	-0.003	0.104	-0.174	-0.282	-0.277	-0.035	-0.473	0.429	1.000				
Bid-ask spread (%)	(11)	0.273	0.000	0.058	-0.270	-0.212	-0.222	-0.084	-0.265	0.281	0.428	1.000			
Book-to-market	(12)	0.104	0.001	0.038	0.088	0.089	-0.164	-0.010	-0.227	0.234	0.180	0.132	1.000		
Past 12-month returns	(13)	-0.101	-0.007	-0.031	-0.009	-0.015	0.024	0.028	0.134	-0.240	-0.187	-0.147	-0.284	1.000	
Gross profitability	(14)	-0.029	0.001	-0.030	-0.015	0.025	0.201	0.010	0.033	-0.061	-0.082	-0.054	-0.297	0.083	1.000

(Continued)

(a) Panel B correlations

Table II—Continued

Panel A: Monthly Sample

	Russell 3000					
	<i>N</i>	Mean	<i>SD</i>	Min	Median	Max
Daily stock volatility (%)	418,748	2.610	1.760	0.581	2.120	11.000
Skewness	418,748	0.123	2.490	-8.900	0.120	8.900
LogAbsVR 5 days(5*1 day))	418,748	-1.740	0.902	-3.850	-1.520	-0.482
ETF ownership	418,748	0.028	0.023	0.000	0.023	0.112
Index fund ownership	418,748	0.047	0.025	0.002	0.045	0.120
Active fund ownership	418,748	0.151	0.083	0.002	0.149	0.354
Hedge fund ownership	418,748	0.058	0.070	0.000	0.031	0.354
log(Mktcap (\$m))	418,748	7.120	1.540	4.250	6.900	11.400
lPrice	418,748	0.073	0.083	0.004	0.047	0.556
Amihud ratio	418,748	0.015	0.017	0.000	0.002	0.078
Bid-ask spread (%)	418,748	0.293	0.527	0.000	0.113	3.700
Book-to-market	418,748	0.568	0.414	0.038	0.498	2.260
Past 12-month returns	418,748	0.133	0.485	-0.823	0.089	2.180
Gross profitability	418,748	0.293	0.258	-0.399	0.262	1.120

(Continued)

(b) Panel A summary statistics, part 2

Table II—Continued

Panel C: Variables Used in Return Regressions (Daily Frequency)

	S&P 500					
	<i>N</i>	Mean	<i>SD</i>	Min	Median	Max
Net flows (\$ million)	1,501,051	0.250	40.700	-12.400	0.027	13.300
abs(Net flows) (\$ million)	1,501,051	2.050	40.700	0.000	0.699	13.300
Net flows/Daily volume	1,501,056	0.002	0.027	-0.081	0.001	0.988
abs(Net flows)/Daily volume	1,501,056	0.015	0.022	0.000	0.007	0.111
NBDO size (\$)	1,457,745	317	1813	54	285	3453
DGTW Ret[t] (%)	1,501,056	0.015	1.700	-4.820	-0.013	5.150
DGTW Ret[t, t + 4] (%)	1,501,056	0.070	3.690	-10.000	-0.094	10.800
DGTW Ret[t, t + 9] (%)	1,501,056	0.127	5.070	-13.500	0.018	14.700
DGTW Ret[t, t + 19] (%)	1,501,056	0.209	7.090	-18.400	0.061	20.200
DGTW Ret[t, t + 39] (%)	1,501,056	0.070	10.000	-25.400	0.136	28.500
abs(Net flows)×10 ⁹ /Mkt cap	1,542,880	0.012	0.016	0.000	0.006	0.078
sum(abs(Flow))×10 ⁹ /Mkt cap	1,542,880	0.018	0.020	0.000	0.012	0.096
Intraday price range (bps)	1,542,880	0.028	0.021	0.006	0.022	0.111
Intraday volatility (%)	1,542,880	0.020	0.018	0.005	0.014	0.100
abs(Mispricing) (bps)	1,666,822	0.174	0.147	0.013	0.129	0.859
Net(mispricing) (bps)	1,660,396	-0.004	0.125	-0.499	0.006	0.369
Share lending fee (%)	1,029,618	0.213	1.110	0.000	0.099	72.400

(Continued)

(b) Panel C daily variables

Table II—Continued

Panel C: Variables Used in Return Regressions (Daily Frequency)

	Russell 3000					
	<i>N</i>	Mean	<i>SD</i>	Min	Median	Max
Net flows (\$ million)	7,812,071	0.081	23.300	-4.600	0.000	5.130
abs(Net flows) (\$ million)	7,812,071	0.617	23.300	0.000	0.091	7.770
Net flows/Daily volume	7,812,107	0.002	0.051	-0.166	0.000	0.176
abs(Net flows)/Daily volume	7,812,107	0.029	0.042	0.000	0.011	0.197
NBDO size (\$)	7,733,982	226	1,057	16	100	1,896
DGTW Ret[t] (%)	7,812,107	0.007	2.070	-5.750	-0.038	6.360
DGTW Ret[t, t + 4] (%)	7,812,107	0.021	4.410	-11.800	-0.095	13.000
DGTW Ret[t, t + 9] (%)	7,812,107	0.026	6.090	-15.800	-0.138	17.400
DGTW Ret[t, t + 19] (%)	7,812,107	0.021	8.540	-21.600	-0.206	23.800
DGTW Ret[t, t + 39] (%)	7,812,107	0.007	12.100	-29.900	-0.359	33.900
abs(Net flows)×10 ⁹ /Mkt cap	8,546,816	0.017	0.022	0.000	0.008	0.094
sum(abs(Flow))×10 ⁹ /Mkt cap	8,546,816	0.022	0.026	0.000	0.013	0.108
Intraday price range (%)	8,546,297	0.036	0.028	0.006	0.028	0.143
Intraday volatility (%)	8,546,816	0.024	0.019	0.005	0.018	0.102
abs(Mispricing) (bps)	8,572,503	0.211	0.169	0.013	0.162	0.859
Net(mispricing) (bps)	8,454,831	-0.015	0.153	-0.499	-0.001	0.369
Share lending fee (%)	4,554,404	0.444	2.240	0.000	0.132	132.000

Panel D: Ancerno Data

	S&P 500					
	<i>N</i>	Mean	<i>SD</i>	Min	Median	Max
Price impact (bps)	28,291,346	3.757	98.373	-463.063	1.242	489.987
Turnover (%)	28,291,346	0.446	1.689	0.000	0.031	24.758

Russell 3000

Price impact (bps)	54,250,928	5.355	114.126	-463.063	1.685	489.987
Turnover (%)	54,250,928	0.939	2.676	0.000	0.077	24.758

5.3 Figures 1 and 2: Two mechanisms for ETF effects

Read:

- Figure 1 illustrates the liquidity-trading mechanism: a demand shock moves the ETF price, arbitrageurs trade the underlying basket, and the stock prices temporarily inherit the shock.

- Figure 2 illustrates the price-discovery mechanism: a fundamental shock is incorporated into ETF prices first, and underlying stocks catch up later.
- The two hypotheses can both produce higher volatility, but they differ in return reversals.

Economic message. The key empirical question is not whether ETF ownership raises volatility alone, but whether that volatility reflects noise or faster information aggregation.

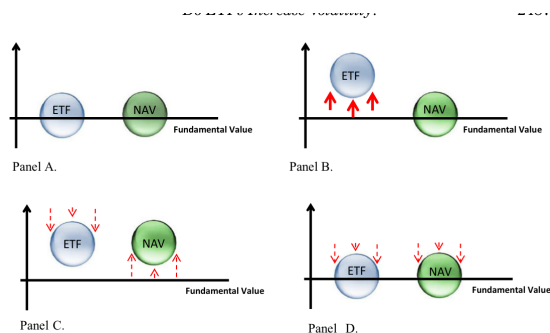


Figure 1. Illustration of the propagation of liquidity shocks via arbitrage. (Panel A) Initial equilibrium. (Panel B) Liquidity shock to ETF. (Panel C) Initial outcome of arbitrage: the shock is propagated to the NAV, and the ETF price starts reverting to the fundamental value. (Panel D) Equilibrium reestablished: after some time, both the ETF price and the NAV revert to the fundamental value. (Color figure can be viewed at wileyonlinelibrary.com)

in Figure 1, Panel C). Thus, ETFs can propagate mispricing to the underlying securities not only because they are traded directly by uninformed investors.

(a) Liquidity-shock propagation

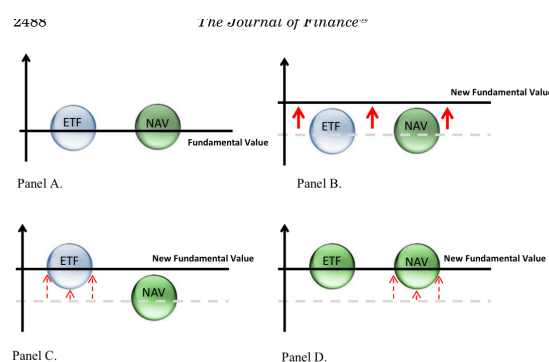


Figure 2. Illustration of the propagation of a fundamental shock with price discovery occurring in the ETF market. (Panel A) Initial equilibrium. (Panel B) Shock to fundamental value. (Panel C) Price discovery takes place in the ETF market. The ETF price moves to the new fundamental value. (Panel D) After a delay, the NAV catches up with the new fundamental. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Price-discovery propagation

5.4 Table III: ETFs are more liquid and attract high-turnover investors

Read:

- ETF bid-ask spreads and Amihud price impact are lower than those of the underlying stocks.
- ETF turnover is much higher than stock turnover.
- Institutional churn ratios are higher for ETF holdings than for stock holdings across institution types.
- Ancerno regressions show that a security being an ETF raises raw and adjusted churn ratios substantially.

Economic message. The clientele premise is supported: ETFs attract short-horizon traders, making them plausible conduits for high-frequency liquidity shocks.

Table III

ETFs versus Stocks: Liquidity, Institutional Ownership, Churn Ratio

The table reports statistics for ETF- and stock-level liquidity, investor turnover, and ownership. The sample includes S&P 500 stocks and ETFs. Panel A reports the security-level liquidity measures (bid-ask spread, the Amihud (2002) ratio, and daily turnover). For all ETFs in our sample, we compute the average measure of liquidity across the stocks in the basket in a given quarter. We then value-weight the ETF- and basket-level measures across all ETFs at the quarter level using ETF market capitalization. Panel B presents information about institutional ownership averaged across all 64 quarters, in the first quarter of the sample, and in the last quarter of the sample. Ownership averages are presented for the ETFs in our sample and all of the stocks in the CRSP data set. Churn ratio is computed as the sum of quarterly absolute changes in dollar holdings over average AUM. Panel C uses trade-level data from Ancerno, aggregated at the manager-stock-quarter level to present regressions of the churn ratio and adjusted churn ratio on an ETF indicator. Summary statistics for Panel C are presented in Table IA.III in the Internet Appendix. *t*-statistics for the test of the null hypothesis that the difference is equal to zero are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The sample covers the period Q1:2000 to Q4:2015.

Panel A: Liquidity and Investors' Churn Ratio Measures							
Liquidity Measures	Quarters	ETFs	Stocks	Difference	<i>t</i> -Statistics		
Bid-ask spread	64	0.002	0.004	-0.002	(-3.611)		
Amihud ratio	64	0.002	0.007	-0.004	(-7.408)		
Daily turnover	64	0.085	0.010	0.075	-13.682		

Panel B: Types of Institutional Ownership							
Type of Institution	Common Stocks				ETFs		
	Ownership				Ownership		
	Total	Q1:2000	Q4:2015	Churn Ratio	Average	Q1:2000	Q4:2015
Bank and trust	1.7%	1.5%	1.9%	17.9%	3.3%	2.5%	4.8%
Insurance company	0.8%	0.8%	0.9%	11.3%	0.5%	1.1%	0.4%
Investment advisor/investment company	38.9%	32.6%	43.0%	18.5%	16.1%	14.2%	25.2%
Investment advisor/hedge fund	14.7%	11.0%	15.6%	26.7%	8.7%	13.7%	6.3%
Hedge fund	2.8%	0.9%	4.2%	74.8%	1.9%	1.0%	0.7%
Pension fund	3.7%	2.7%	3.4%	13.3%	1.4%	1.6%	1.1%
Research firm	2.0%	0.8%	2.9%	39.7%	17.6%	7.4%	17.6%
Other institutions	0.7%	0.4%	0.7%	23.7%	1.3%	0.8%	0.1%
All institutions	65.4%	50.9%	72.4%	22.6%	50.7%	42.2%	56.3%

Panel C: Churn Ratio Regressions						
Dependent Variable:	Churn Ratio			Adjusted Churn Ratio		
	(1)	(2)	(3)	(4)	(5)	(6)
Security is ETF	0.194*** (13.764)	0.197*** (14.584)	0.146*** (11.060)	0.571*** (10.258)	0.570*** (10.142)	0.372*** (7.246)
Date fixed effects	No	Yes	Yes	No	Yes	Yes
Manager fixed effects	No	No	Yes	No	No	Yes
Observations	32,371	32,371	32,329	32,371	32,371	32,329
Adjusted- <i>R</i> ²	0.154	0.168	0.358	0.098	0.101	0.396

5.5 Table IV and Figure 3: OLS volatility and Russell cutoff

Read:

- Table IV shows that ETF ownership is positively related to stock volatility in OLS.
- In the S&P 500, a one-standard-deviation increase in ETF ownership is associated with about 0.164 standard deviations higher daily volatility in the baseline specification.
- The coefficient remains positive after adding other institutional ownership and lagged volatility.
- Figure 3 shows the discontinuity in ETF ownership around the Russell cutoff; stocks moving into the Russell 2000 receive higher ETF ownership.

Economic message. OLS gives the broad correlation; the Russell cutoff motivates the IV strategy by showing an exogenous-looking jump in ETF ownership.

Table IV – Continued

Dependent Variable	Panel A: OLS Regressions, Post-Index									
	Daily Volatility (1)					Daily Volatility (2)				
Sample	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ETF ownership	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***
logMktcap (1 - 1)	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001
12Month (1 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
12Month (2 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Active fund ownership (1 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Active fund ownership (2 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Market fund effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771
R ²	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008

(a) Table IV, part 1

Table IV – Continued

Dependent Variable	Panel B: OLS Regressions, Post-Index									
	Daily Volatility (1)					Daily Volatility (2)				
Sample	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ETF ownership	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***
logMktcap (1 - 1)	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001
12Month (1 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
12Month (2 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Active fund ownership (1 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Active fund ownership (2 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Market fund effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771
R ²	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008

(b) Table IV, part 2

(c) Figure 3: Russell cutoff

Figure 3: Fund ownership around the Russell cutoff. Top: Average ownership (in %) by ETFs (Panel A), index funds (Panel B), active funds (Panel C), and hedge funds (Panel D) for stocks ranked by market capitalization and included in the Russell 2000. The average is computed first by ranking over time, and then ranking across time of 10 stocks. The vertical line denotes the 1,000th stock. The sample covers the period January 2000 to May 2007. (Color figure can be viewed at [www.intellectus.com](#))

Table IV – Continued

Dependent Variable	Panel B: OLS Regressions, Sample Split by Period									
	2000 to 2006					2007 to 2008				
Sample	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ETF ownership	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***
logMktcap (1 - 1)	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001
12Month (1 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
12Month (2 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Active fund ownership (1 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Active fund ownership (2 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Market fund effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771
R ²	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008

(a) Table IV, sample split

Table V
Quasi-Natural Experiment Based on the Russell Index Reconstitution

The table reports estimates from a quasi-natural experiment that relies on the reconstitution of the Russell 1000 and Russell 2000 indexes. The sample is at the stock-month level. Panel A presents the first-stage results where the dependent variable is ETF ownership. The explanatory variables (instruments) are a dummy for inclusion in the Russell 2000, for stocks in the Russell 1000 before index reconstitution (columns (1) to (5)), and a dummy for inclusion in the Russell 1000, for stocks in the Russell 2000 before index reconstitution (columns (6) to (10)). Stocks are ranked by market capitalization in May of each year. Columns (1) to (5) and (6) to (10) present bandwidths ranging from 100 to 500 stocks. The same stocks enter the sample in June after index reconstitution and remain in the sample until the following May, except if delisted. The controls in all panels are lagged market capitalization, lagged average share price, lagged Amihud (2002) ratio, lagged average bid-ask spread, lagged book-to-market ratio, lagged past 12-month returns, lagged gross profitability (as in Newey-Marc (2013)), lagged volatility, index fund ownership, active fund ownership, hedge fund ownership, and a linear specification of the ranking variable (not reported). In Panels B, C, and D, the dependent variable is daily stock volatility (computed using daily returns within a month). The main explanatory variable is instrumented ETF ownership. The instruments are the switching indicators. In Panel C, the main explanatory variable is an interaction between the dummy variables for index inclusion and average ETF ownership (for Russell 2000 stock). Panel D also includes controls for average ETF ownership as well as average ownership for index funds, active funds, and hedge funds (for Russell 2000 stock). In Panel D, the main explanatory variable is an interaction between the dummy variable for Russell 2000 inclusion and the stock-level ratio of ETF ownership to other fund (active, active, and hedge fund) ownership in May before reconstitution. The dependent variable and the control variables are standardized by subtracting the mean and dividing by the standard deviation. Standard errors are double-clustered at the stock and month levels. t -statistics are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The sample covers the period June 2000 to May 2007.

Dependent Variable	Panel A: First-Stage Regressions, First Stage Post-Index									
	Switch to the Russell 2000					Switch to the Russell 1000				
Bandwidth	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Switch indicator	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***	0.180***
logMktcap (1 - 1)	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001
12Month (1 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Active fund ownership (1 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Active fund ownership (2 - 1)	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Market fund effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771	20,771
R ²	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008	0.008

(b) Table V first stage

5.6 Table V: Russell reconstitution IV

Read:

- The first stage is strong: switching to the Russell 2000 raises ETF ownership, while switching to the Russell 1000 lowers it.
- The second stage finds positive effects of instrumented ETF ownership on volatility across bandwidths.
- The paper reports effects ranging from roughly 0.169 to 0.717 standard deviations, averaging about 0.31.
- Additional tests interact index switches with aggregate ETF ownership and with the ratio of ETF ownership to other institutional ownership.

Economic message. The IV evidence supports a causal interpretation: the ETF ownership component induced by index assignment raises stock volatility.

5.8 Table VII: ETF flows and return reversals

Read:

- ETF flows create contemporaneous price pressure in the underlying stocks.
- A one-standard-deviation stock-level ETF flow is associated with roughly 12 bps same-day price movement for S&P 500 stocks.
- Much of the price pressure reverses over the following 40 trading days.
- Ancerno price-impact estimates support the magnitude of the flow-induced price impact.

Economic message. Return reversal is the main evidence that ETF-induced volatility is not simply price discovery. It is consistent with temporary nonfundamental demand shocks.

Table VII
ETF Flows, Stock Returns, and Price Impact

Panel A DOTW-adjusted cumulative returns (Daniel et al., 1997) over different horizons starting on day t on stock-level ETF flows on day t . The analysis is at the stock-day level. ETF flows are scaled by market capitalization and standardized. The controls in all panels include lagged market capitalization, the lagged inverse share price, the lagged Amihud (2002) ratio, lagged average bid-ask spread, lagged book-to-market ratio, lagged past 12-month returns, lagged gross profitability (as in Nov-Meyers (2013)), stock-level order imbalance, and lagged returns. Stock and quarter fixed effects are included in Panel A, and quarter fixed effects are included in Panel B. Standard errors are double-clustered at the stock and time levels. Panel B reports estimates from ordinary least squares (OLS) regressions of price impact on turnover and turnover squared. The sample includes trade-level observations from Ancerno data. Price impact is defined as the percentage difference between the execution price and the price at time of placement. Turnover is measured as dollar volume of a trade as a fraction of daily dollar volume in CBOB. Day and stock fixed effects are included. Standard errors are double-clustered at the stock and time levels. The sample covers the period January 2000 to December 2014. t -statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Sample:	Panel A: ETF Flows and Stock Returns									
	S&P 500					Russell 2000				
Dependent Variable:	Ret(t)	Ret($t+4$)	Ret($t+9$)	Ret($t+19$)	Ret($t+39$)	Ret(t)	Ret($t+4$)	Ret($t+9$)	Ret($t+19$)	Ret($t+39$)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
ETF flows (t)	0.122***	0.111***	0.062***	0.046**	0.015	0.040***	0.034***	0.027***	0.016*	-0.009
logMktcap ($t-1$)	(13.433)	(7.469)	(3.233)	(1.848)	(0.436)	(15.069)	(6.649)	(3.995)	(1.698)	(-0.096)
logMktcap ($t-10$)	0.048***	0.039***	0.023**	0.040	0.050	0.008***	0.027***	0.044***	0.009***	0.311***
(14.352)	(4.750)	(2.093)	(1.216)	(0.738)	(0.956)	(7.371)	(7.942)	(8.579)	(8.896)	
1Pctret ($t-1$)	-0.001***	0.009	1.471***	4.531***	8.568***	-0.316***	0.100	0.612***	1.439***	4.754***
(-4.084)	(0.835)	(2.553)	(3.965)	(3.986)	(-10.093)	(1.302)	(4.369)	(6.757)	(8.791)	
Amihud ($t-1$)	37.342***	1.842	-11.973	-72.013	-139.467	0.137***	-0.113***	-1.569***	-3.664***	-6.602***
(6.713)	(0.139)	(-1.399)	(-1.332)	(-1.482)	(3.777)	(-8.169)	(-19.061)	(-11.816)	(-11.864)	
Bid-ask spread ($t-1$)	-2.459**	2.411	6.613*	10.021***	20.769**	0.718*	0.105	0.261	1.653	4.524
(-2.505)	(1.132)	(1.759)	(2.671)	(2.220)	(1.853)	(0.157)	(0.196)	(0.716)	(1.048)	
Book-to-market ($t-1$)	0.020*	0.014	0.011	-0.001	-0.020	0.017***	0.019**	0.002***	0.107**	0.177**
(1.702)	(0.965)	(0.232)	(-0.013)	(-0.106)	(1.511)	(2.051)	(2.622)	(2.333)	(1.966)	
Past 12-month returns ($t-1$)	-0.014	-0.016	-0.026	0.067	0.308	-0.020***	-0.007	-0.006	0.017	0.043
(-1.591)	(-0.967)	(-0.699)	(0.878)	(1.480)	(-5.525)	(-0.657)	(-0.306)	(0.403)	(0.561)	
Gross profitability ($t-1$)	0.015**	0.053	0.045	0.064	0.152	0.049***	0.222***	0.403***	0.499***	1.210***
(2.052)	(1.204)	(0.627)	(0.481)	(0.671)	(6.435)	(9.440)	(10.277)	(9.916)	(9.200)	
Order imbalance ($t-1$)	0.003***	0.003***	0.002***	0.002***	0.002***	0.002***	0.002***	0.002***	0.002***	0.002***
(30.993)	(10.913)	(10.541)	(9.655)	(4.933)	(15.636)	(61.030)	(51.332)	(41.312)	(27.833)	

(Continued)

(a) Table VII, flows and returns

Table VII—Continued

Panel A: ETF Flows and Stock Returns

Sample:	Panel A: ETF Flows and Stock Returns									
	S&P 500					Russell 2000				
Dependent Variable:	Ret(t)	Ret($t+4$)	Ret($t+9$)	Ret($t+19$)	Ret($t+39$)	Ret(t)	Ret($t+4$)	Ret($t+9$)	Ret($t+19$)	Ret($t+39$)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Ret($t-1$)	-2.587***	(-8.348)				-3.364***	(-19.580)			
Ret($t-1, t-5$)			-2.109**	(-7.195)				-1.571***	(-10.302)	
Ret($t-1, t-10$)					-3.388***	(-7.307)				-1.981***
Ret($t-1, t-20$)						-2.496***	(-7.690)			-1.973***
Ret($t-1, t-40$)								-3.068***	(-4.240)	-1.409***
Day fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,492,313	1,492,313	1,492,313	1,492,313	1,492,313	7,708,143	7,708,143	7,708,143	7,708,143	7,708,143
R ²	0.033	0.013	0.011	0.011	0.011	0.013	0.039	0.009	0.007	0.006

Panel B: Price Impact on Turnover

Sample:	Price Impact			
	S&P 500		Russell 2000	
	(1)	(2)	(3)	(4)
Turnover (%)	3.609***	3.621***	2.426***	3.465***
(34.627)		(36.255)	(47.787)	(41.892)
Turnover (%) ²	-0.148***	-0.148***	-0.141***	-0.131***
(-39.961)		(-39.910)	(-38.559)	(-35.661)
Constant	2.696***	2.696***	3.199***	3.199***
(15.509)		(15.509)	(15.126)	(15.126)
Day fixed effects	No	Yes	No	Yes
Stock fixed effects	No	Yes	No	Yes
Observations	27,960,051	27,960,051	53,167,792	53,167,792
R ²	0.001	0.009	0.002	0.011

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5.9 Table VIII: Limits to arbitrage

Read:

- Absolute ETF mispricing predicts higher next-day intraday stock volatility.
- The effect is weaker for stocks with high bid-ask spreads.
- The effect is also weaker when lending fees are high, especially for negative mispricing episodes that require shorting.

Economic message. The volatility effect operates through arbitrage. When arbitrage is expensive, ETF mispricing is less strongly transmitted to the underlying stock.

Table X
ETF Ownership Portfolios

The table reports statistics and regression analysis for portfolios based on ETF ownership. Each month, five equally weighted portfolios are formed on the basis of the five quintiles of the distribution of ETF ownership in the previous month. Panel A presents the raw returns. Panel B presents factor model regressions for the high-ETF-ownership minus the low-ETF-ownership portfolio. The factors are the five Fama and French (2015) factors (MKTRF, HML, SMB, RMW, CMA), momentum (UMD), and the Pastor and Stambaugh (2003) traded liquidity factor (PS.VWF). Panel C reports estimates from Fama and MacBeth (1973) regressions of four-factor model-adjusted (Carhart (1997)) monthly stock returns on ETF ownership in the previous month, contemporaneous ETF stock-level flows, index, active, and hedge fund ownership, and control variables. The dependent variable is expressed as a percentage, and both the ownership variables and flows are standardized. The controls include logged market capitalization, the lagged inverse share price ratio, the lagged Amihud (2002) ratio, the lagged average bid-ask spread, the lagged book-to-market ratio, lagged past 12-month returns, and lagged gross profitability (as in Novy-Marx (2013)). *t*-statistics are presented in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The sample covers the period January 2000 to December 2015.

Panel A: Raw Excess Returns for the Quintile Portfolios								
	Quintiles Based on ETF Ownership							
	Low	(2)	(3)	(4)	High			
Raw excess returns	0.389 (1.051)	0.618* (1.736)	0.696* (1.941)	0.739** (2.040)	0.737* (1.828)			
Number of months	191	191	191	191	191			
Panel B: High-Minus-Low Portfolio, Full Sample								
Dependent Variable:	Ret(High-Minus-Low ETF Ownership)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Alpha	0.347** (2.296)	0.303** (2.074)	0.334** (2.294)	0.355** (2.429)	0.364** (2.477)	0.350** (2.287)	0.383** (2.482)	0.383** (2.461)
MKTRF		0.130*** (4.012)	0.126*** (3.915)	0.136*** (4.126)	0.125*** (3.437)	0.132*** (3.204)	0.119*** (2.847)	0.119*** (2.829)
HML			-0.089** (-1.996)	-0.095** (-2.127)	-0.101** (-2.223)	-0.111** (-2.046)	-0.051 (-0.749)	-0.051 (-0.738)
SMB				-0.061 (-1.316)	-0.054 (-1.132)	-0.044 (-0.778)	-0.036 (-0.645)	-0.036 (-0.620)
UMD					-0.021 (-0.747)	-0.023 (-0.787)	-0.014 (-0.475)	-0.014 (-0.474)
RMW						0.025 (0.340)	0.016 (0.220)	0.016 (0.219)
CMA							-0.136 (-1.460)	-0.136 (-1.455)
PS.VWF								-0.001 (-0.021)
Number of months	191	191	191	191	191	191	191	191
R ²	0.000	0.078	0.098	0.106	0.109	0.109	0.119	0.119

(Continued)

Table X—Continued

Dependent Variable:	Panel C: Fama and MacBeth (1973) Regressions				
	Four-Factor Model-Adjusted Returns (<i>t</i>)				
	(1)	(2)	(3)	(4)	(5)
ETF ownership (<i>t</i> − 1)	0.144** (2.382)	0.161** (2.536)	0.195*** (2.862)		0.199*** (2.642)
ETF flows (<i>t</i>)				0.099** (2.009)	0.080 (1.452)
Hedge fund ownership (<i>t</i> − 1)		0.062 (1.198)	0.024 (0.444)	0.026 (0.474)	0.021 (0.376)
Index fund ownership (<i>t</i> − 1)		-0.055 (-1.145)	-0.065 (-1.363)	-0.025 (-0.575)	-0.067 (-1.398)
Active fund ownership (<i>t</i> − 1)		0.075 (1.231)	0.049 (0.973)	0.053 (1.048)	0.040 (0.819)
log(Mktcap (<i>t</i> − 1))			0.109* (1.895)	0.029 (0.450)	0.078 (1.310)
1/Price (<i>t</i> − 1)			6.458*** (5.197)	6.440*** (5.209)	6.206*** (5.011)
Amihud (<i>t</i> − 1)			3.074 (0.182)	-10.612 (-0.607)	6.454 (0.364)
Bid-ask spread (<i>t</i> − 1)			-18.181 (-1.281)	-21.054 (-1.479)	-17.810 (-1.254)
Book-to-market (<i>t</i> − 1)			-0.068 (-0.311)	-0.095 (-0.440)	-0.067 (-0.304)
Past 12-month returns (<i>t</i> − 1)			-0.430 (-1.495)	-0.408 (-1.442)	-0.417 (-1.453)
Gross profitability (<i>t</i> − 1)			0.709** (2.332)	0.659** (2.181)	0.702** (2.321)
Observations	350,237	350,237	350,237	350,237	350,237
R ²	0.003	0.011	0.037	0.036	0.039
Number of months	191	191	191	191	191

the first specification reveals a positive and significant relation between returns and standardized ETF ownership. A one-standard-deviation increase in ETF ownership correlates with a 14 bps increase in returns. The next specification includes ownership by other types of institutions, whose effect is not significant, although it is positive in the case of active mutual funds and hedge funds. We also add controls for liquidity and stock characteristics typically associated with returns. We note a significant profitability effect, consistent with Novy-Marx (2013), whereas the other asset pricing effects are insignificant, probably due to the four-factor adjustment of the dependent variable. In column (4), ETF flows display a significant correlation with returns when ETF ownership is not included. However, a horse race between the two variables reveals that ETF ownership is the dominant determinant of stock returns (column (5)). This finding rules out the possibility that the alphas we identify are due to the price pressure of ETF flows.

Overall, these results support the argument that ETFs significantly modify the underlying stocks' return distribution. As ETFs increase the volatility of stock returns, they appear to introduce a nondiversifiable source of risk, at

(a) Table X, portfolio returns and alphas

(b) Table X, Fama–MacBeth regressions

6 Short summary

- The paper studies whether ETFs increase the volatility of their underlying stocks.
- The hypothesized channel is liquidity-shock propagation through ETF arbitrage.
- ETFs are more liquid than underlying stocks and attract higher-turnover investors.
- ETF ownership is positively related to daily stock volatility in OLS.
- The Russell 1000/2000 reconstitution provides quasi-exogenous variation in ETF ownership.
- The IV results show that ETF-induced ownership changes raise stock volatility.
- ETF ownership also raises the churn ratio of traders in the underlying stocks.
- ETF flows generate same-day price pressure that subsequently reverses.
- Variance-ratio tests show greater deviation from a random walk and more negative autocorrelation.
- The arbitrage-channel evidence is strongest when ETF mispricing is high and arbitrage is not too costly.
- High-ETF-ownership stocks earn positive return premia.
- The overall conclusion is that ETFs can add nonfundamental volatility and systematic risk to the underlying securities.

7 Conclusion

7.1 Core conclusion

Ben-David, Franzoni, and Moussawi show that ETF ownership is associated with higher stock-level volatility and provide quasi-experimental evidence that the relation is causal. The mechanism is not simply that ETFs hold volatile stocks. Rather, ETF trading and arbitrage transmit short-horizon liquidity demand into the underlying stocks.

7.2 Economic intuition

The economic intuition is that ETFs create a very cheap trading venue for baskets. Investors trade the ETF for liquidity, hedging, or short-horizon reasons. When the ETF price deviates from its NAV, arbitrageurs trade the component stocks. This mechanical arbitrage makes the component stocks inherit the ETF-level demand shock. Because the shock is at least partly temporary, prices later mean revert.

7.3 Takeaway

ETFs improve access to diversification and liquidity, but they also create a channel through which nonfundamental demand shocks propagate to individual stocks. ETF ownership is therefore not only an ownership statistic; it is a measure of exposure to a trading technology that can affect volatility, return autocorrelation, and expected returns.